

Alex Duda

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EDUCATION

McGill University, Faculty of Arts

Sept 2024 to Dec 2027 (Expected)

Bachelor of Arts, Major in Computer Science, Minor in Anthropology

Relevant Coursework: Calculus I to III, Linear Algebra I to II, Discrete Mathematics, Probability, Intro to Computer Science, Software Systems, Computer Systems, Algorithms and Data Structures, Theory of Computation, Paradigms and Languages

SOFTWARE PROJECT EXPERIENCE

Symbolos: AI-Powered Academic Advising Platform

Jul 2025 to Present

Full-stack web app. Python, FastAPI, React/Vite, Supabase (PostgreSQL), Claude API, GitHub Actions CI/CD

- Designed, built, and deployed a production full-stack AI advising platform, free for every McGill student (FastAPI backend on Vercel, React/Vite frontend, Supabase for auth and data), integrating the Claude API for contextual, LLM-driven academic guidance
- Engineered a black-box application security testing suite (Nuclei scanners, stateful Python probes for JWT/RLS/IDOR vulnerabilities, and a GitHub Actions CI/CD pipeline splitting fast and slow checks), identifying and resolving 25 flagged vulnerabilities across four audit sweeps
- Localized the product to English, French, and Mandarin (300+ translation keys) and manage a team of 5 (engineering, design, growth) to drive engagement through a technical SEO/GEO strategy (schema markup, llms.txt, community outreach), and, without any paid marketing, reached 70 users
- Reached out and met with artists around North America to discuss and plan sponsoring an affordable concert for McGill students

AI Auditor: General-Purpose LLM Output Auditing Pipeline

Jul 2026 to Present

CLI tool. Python, Anthropic SDK (Claude), concurrent pipeline architecture, code execution and web search tool use

- Architected a multi-stage audit pipeline that verifies any LLM-generated solution by extracting and testing its underlying assumptions, running executable checks via a code-execution tool, searching the web for counter-evidence, and computing a calibrated confidence score with an explanation of remaining uncertainty
- Ran independent, mutually exclusive pipeline stages concurrently to cut wall-clock latency, and implemented explicit error handling so model refusals or truncated (max-token) responses raise an error instead of silently producing incomplete output

Plumb (@plumb/scanner): Open-Source Codebase Intelligence Tool

Jun 2026 to Present

VS Code extension and MCP server. TypeScript, Node.js, YAML

- Built and published an open-source VS Code extension and MCP server that scans GitHub repositories and recommends verified open-source solutions, scored by a custom six-signal automated evaluation engine (Assay)
- Seeded and maintain a community-driven YAML registry spanning 20 entries across seven categories

Tokenized Treasury Dashboard

Jul 2026 to Present

Data pipeline and web app. Python, SQLAlchemy/SQLite, TypeScript, React, Vite, Recharts, GitHub Actions

- Built a full-stack data pipeline tracking \$9.8B+ across six tokenized U.S. Treasury products (BUIDL, USYC, OUSG, and others) on 8+ blockchains, ingesting 18,000+ historical snapshots from the DefiLlama API into a normalized SQLite schema
- Implemented on-chain verification (direct smart-contract reads via public RPC) to cross-check reported figures against ground-truth token supply, with fully automated daily ingestion and deployment to GitHub Pages via GitHub Actions

GrantGuard: Corruption-Resistant Grant Allocation Algorithm

May 2026 to Present

Algorithm design and simulation. Python, Monte Carlo methods

- Designed a corruption-resistant algorithm for government grant and contract allocation across three design iterations, informed by empirical analysis of real-world corruption signatures across multiple public datasets
- Built a weighted Corruption Indicator Matrix and a Monte Carlo simulation framework to stress-test allocation fairness under adversarial conditions

CERTIFICATIONS

Google Data Analytics Professional Certificate

Aug 2025 to Dec 2025 (Expected)

- Completed 8 courses covering data cleaning, transformation, analysis, and visualization in SQL, R, Excel, and Tableau; applied descriptive statistics, correlation analysis, and hypothesis testing to real-world datasets in a hands-on capstone project

PROFESSIONAL EXPERIENCE

Harvard Medical School and Massachusetts General Hospital, Student Intern

Mar 2023 to Aug 2023

- Assisted with experimental research on vessel formation, co-cultures, cancer cell growth, and pancreatic cancer treatments; collected, analyzed, and interpreted experimental data to support research conclusions

Stone Hearth Pizza Co., Cashier and Server

Jul 2021 to Aug 2024

- Managed customer service and coordinated food delivery and take-out operations for 200+ customers daily; earned a 10% raise for taking on expanded responsibilities

LEADERSHIP AND EXTRACURRICULAR

D1 High School Varsity Soccer: Captain, Center Back

Aug 2022 to Dec 2024

- Led team to a top-3 league finish and state tournament quarterfinals (voted Most Improved Player); earned captaincy mid-season after proving myself as a leader and helped orchestrate the largest upset in tournament history through practice, planning, and performance analysis

SKILLS AND INTERESTS

Languages & Tools: Java, Python, TypeScript/JavaScript, C, SQL, HTML/CSS

Frameworks & Platforms: React, React Native, Node.js, FastAPI, Supabase/PostgreSQL, SQLite, Git, GitHub Actions (CI/CD), Linux/Unix, REST APIs, Pydantic, Homebrew, Docker

AI & Data: LLM integration (Anthropic Claude API), prompt engineering, web scraping, data analysis (R, Excel, Tableau)

Spoken Languages: Fluent English, Intermediate French and Romanian, Basic Italian

Interests: Brentford FC, Travelling and Eating, Data Privacy